United States Patent	12409461
Kind Code	B2
Date of Patent	September 09, 2025
Inventor(s)	Hasegawa; Takaaki et al.

Dust collector and dust collection method

Abstract

Provides a dust collector capable of being installed even in small spaces and capable of collecting dust efficiently. A dust collector **1** includes a first electrode layer **12** formed of a conductor, a second electrode layer **14** formed of a conductor, having sufficient size to cover the first electrode layer **12** in planar view, having lacking parts (through holes **14**H, slits **14**S) piercing therethrough in a thickness direction, having sufficient size to cover the first electrode layer **12** in planar view, and placed interfacing the first electrode layer **12**, and an insulating layer (a first insulating layer **13**, a second insulating layer **15**) formed of an insulative material, insulating the first electrode layer **12** and the second electrode layer **14**, and forming a single merged layer structure with the first electrode layer **12** and the second electrode layer **14.**

Inventors:	**Hasegawa; Takaaki (Kawasaki Kanagawa, JP), Moriyama; Hiroyuki (Kawasaki Kanagawa, JP)**
Applicant:	**Creative Technology Corporation** (Kanagawa, JP)
Family ID:	**80846473**
Assignee:	**Creative Technology Corporation (Kanagawa, JP)**
Appl. No.:	**17/768465**
Filed (or PCT Filed):	**August 31, 2021**
PCT No.:	**PCT/JP2021/031967**
PCT Pub. No.:	**WO2022/064977**
PCT Pub. Date:	March 31, 2022

Prior Publication Data

Document Identifier	**Publication Date**
US 20230381791 A1	Nov. 30, 2023

Foreign Application Priority Data

JP	2020-159465	Sep. 24, 2020

Publication Classification

Int. Cl.: **B01D53/22** (20060101); **B03C3/47** (20060101); **B03C3/64** (20060101); **B03C3/70** (20060101)

U.S. Cl.:

CPC **B03C3/70** (20130101); **B03C3/47** (20130101); **B03C3/64** (20130101);

Field of Classification Search

CPC: B03C (3/70); B03C (3/47); B03C (3/64)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2864460	12/1957	Powers	55/DIG.1	B03C 3/155
2917130	12/1958	Powers	96/67	B01D 46/0032
4750921	12/1987	Sugita	96/67	B03C 3/155
4798575	12/1988	Siversson	493/347	B65H 45/20
5234772	12/1992	Oguchi	428/473.5	G02B 6/12007
10357781	12/2018	Luo	N/A	N/A
2004/0216745	12/2003	Yuen	128/205.27	A62B 19/00
2006/0000197	12/2005	Haefner	55/521	B01D 46/521
2007/0223173	12/2006	Fujisawa et al.	N/A	N/A
2008/0047434	12/2007	Kobayashi et al.	N/A	N/A
2008/0251718	12/2007	Kaga et al.	N/A	N/A
2008/0276805	12/2007	Lotgerink-Bruinenberg	96/75	B60H 3/0658
2012/0074316	12/2011	Watanabe et al.	N/A	N/A
2012/0120545	12/2011	Fujisawa et al.	N/A	N/A
2016/0207051	12/2015	Tatsumi et al.	N/A	N/A
2017/0120181	12/2016	Jeon	N/A	F24F 13/20
2018/0226392	12/2017	Lee	N/A	H01L 23/60
2020/0096881	12/2019	Van Duivenbode et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
106268085	12/2016	CN	B01D 46/023
H09-262500	12/1996	JP	N/A
2001-179128	12/2000	JP	N/A
2004-298660	12/2003	JP	N/A
2009-004161	12/2008	JP	N/A
2012-064567	12/2011	JP	N/A
5500172	12/2013	JP	N/A
6620994	12/2018	JP	N/A
2020-522020	12/2019	JP	N/A
20180104207	12/2017	KR	N/A
2005/091356	12/2004	WO	N/A
2015-029698	12/2014	WO	N/A

OTHER PUBLICATIONS

"Planar view" American Heritage Dictionary of the English Language, Fifth Edition. 1 page. 2016 by Houghton Mifflin Harcourt Publishin Company <https://www.thefreedictionary.com/planar> (Year: 2016). cited by examiner
"Layer" American Heritage Dictionary of the English Language, Fifth Edition. 1 page. 2016 by Houghton Mifflin Harcourt Publishing Company. <https://www.thefreedictionary.com/layer> (Year: 2016). cited by examiner
"Electric fields—Forces on charged particles—Higher Physics Revision—BBC Bitesize" https://www.bbc.co.uk/bitesize/guides/zyfyyrd/revision/2#:˜:text=In%20an%20electric%20field%20a,it%20will%20push%20them%20apar retrieved 2024, 5 pages (Year: 2024). cited by examiner
Machine translation of Dae K S CN-106268085-A Jan. 4, 2017 (Year: 2017). cited by examiner
"Slit", American Heritage Dictionary of the English Language, Fifth Edition. 2016 by Houghton Mifflin Harcourt Publishing Company. 1 page https://www.thefreedictionary.com/slit (Year: 2016). cited by examiner
Translation of the International Search Report mailed Nov. 2, 2021 in International Application No. PCT/JP2021/031967, 14 pages. cited by applicant

---

Primary Examiner: Shumate; Anthony R

Attorney, Agent or Firm: Kim & Stewart LLP

Background/Summary

TECHNICAL FIELD
(1) The present invention relates to a dust collector and a dust collection method.
BACKGROUND TECHNOLOGY
(2) For example, fine dust (particles) generated within or near manufacturing equipment such as in semiconductor device fabrication tools is often problematic, contributing to contamination defects on the processed workpiece. Therefore, there exists a need to reduce particles and maintain cleanliness near the workpiece, within or in the vicinity surrounding the manufacturing equipment.
(3) Regarding this conception, methods such as a filter dust collection system, an electric dust collection system, and an electric-

field curtain system have been previously proposed.
(4) In the filter dust collection system, dispersed airborne particles are drawn via an air duct, and the air is purified by the air filter. Consequently, it is not possible to collect particles prior to its dispersal, and the high pressure drop yields to a low dust collection efficiency.
(5) In the electric dust collection system, particles are charged through a corona discharge by an ionizer, and the charged particles are collected to its opposing-charged bipolar electrode. Accordingly, installation locations of the collection system are restricted due to the ionization of the peripheral environment and the enlarged footprint.
(6) A dust collection device having no ionizer (for example, Patent Literature 1) has been also proposed; however, since air must pass between the bipolar electrode structure, space saving has its limitations.
(7) In the electric-field curtain system, unbalanced electric fields, traveling in a single direction are generated by applying alternating current to polyphase linear electrodes buried in a dielectric. Then, charged particles are caused to float and is pushed towards the direction to be collected. Likewise, due to its large size, installation locations of the collection system are restricted.
CITATION LIST
Patent Literature
(8) PTL 1: Japanese Patent No. 6620994
SUMMARY OF INVENTION
Technical Problem
(9) From above observations, a dust collector and a dust collection method are demanded of which can be installed even in small spaces and efficiently collect dust.
Solution to Problem
(10) The present invention provides a dust collector including a first electrode layer consisted by a conductor, a second electrode layer consisted by a conductor, of which having lacking parts therethrough in the thickness direction, having sufficient size to cover the first electrode layer in planar view, constructed to interface the first electrode layer, and an insulating layer consisting of an insulative material, insulating the first electrode layer and the second electrode layer, forming a single merged layer structure with the first electrode layer and the second electrode layer.
Advantageous Effects of Invention
(11) According to the present invention, it is possible to provide a dust collector and a dust collection method that can be installed even in small spaces of which can collect dust efficiently.

Description

BRIEF DESCRIPTION OF DRAWINGS
(1) FIG. **1** is an external perspective view of the dust collector.
(2) FIG. **2** is a cross-sectional view of the dust collector taken along A-A line in FIG. **1**.
(3) FIG. **3** is a planar view of respective layers of the dust collector.
(4) FIG. **4** is a cross-sectional view showing a state of electric fields generated in the dust collector.
DESCRIPTION OF EMBODIMENTS
(5) Hereinafter, a dust collector according to an embodiment of the present invention will be explained in detail.
(6) FIG. **1** is an external perspective view of a dust collector **1** according to the embodiment, FIG. **2** is a cross-sectional view of the dust collector **1** taken along A-A line in FIG. **1**, and FIG. **3** is a planar view of respective layers of the dust collector **1**.
(7) As shown in FIG. **1** and FIG. **2**, the dust collector **1** includes a support base layer **11**, a first electrode layer **12**, a first insulating layer **13**, a second electrode layer **14**, a second insulating layer **15**, and a dust attraction layer **16**.
(8) A power supply device **2** is connected to the first electrode layer **12** and the second electrode layer **14**.
(9) The support base layer **11** is a layer formed of an insulator, more preferably an insulative synthetic resin.
(10) The first electrode layer **12** is a layer formed of a conductor. Examples of the conductor include a gold leaf, a conductive synthetic resin, and carbon. The first electrode layer is provided on one surface of the support base layer **11**.
(11) The first insulating layer **13** is a layer formed of an insulator, more preferably an insulative synthetic resin not interfering with an electric field. Examples of the insulative synthetic resin not interfering with electric field include polyimide. The first insulating layer **13** is provided on a side of the support base layer **11** where the first electrode layer **12** is provided and surrounds the first electrode layer **12** with the support base layer **11** or alone surrounds the first electrode layer **12**, for insulation.
(12) The second electrode layer **14** is a layer formed of a conductor. Examples of the conductor include a gold leaf, a conductive synthetic resin, and carbon. The second electrode layer **14** is provided at a position where the second electrode layer **14** is opposed to the first electrode layer **12** with the first insulating layer **13** interposed therebetween. The second electrode layer **14** has lacking parts piercing therethrough in a thickness direction and has sufficient size to cover the first electrode layer **12** in planar view.
(13) The second insulating layer **15** is a layer formed of an insulator, more preferably an insulative synthetic resin not interfering with an electric field. For example, polyimide can be cited as the insulative synthetic resin not interfering with the electric field. The second insulating layer **15** is placed at a position where the second insulating layer **15** is opposed to the first insulating layer **13** with the second electrode layer **14** interposed therebetween, and surrounds the second electrode layer **14** with the first insulating layer **13** or alone surrounds the second electrode layer **14**, for insulation.
(14) The dust attraction layer **16** is placed at a position where the dust attraction layer **16** is opposed to the second electrode layer **14** with the second insulating layer **15** interposed therebetween. The dust attraction layer **16** is formed of a material suitable for collecting fine dust (particles) and retaining them. An adhesive, a material with high intermolecular force or a material with high friction force can be suitably selected and adopted as the material for the dust attraction layer **16**.
(15) The dust collector **1** is formed as a layered sheet or a layered plate in which the support base layer **11**, the first electrode layer **12**, the first insulating layer **13**, the second electrode layer **14**, the second insulating layer **15**, and the dust attraction layer **16** are stacked in this order.

(16) The first insulating layer **13** and the second insulating layer **15** may be integrally formed. Moreover, the support base layer **11** may be integrally formed with the first insulating layer **13** and the second insulating layer **15** using the same material as the first insulating layer **13** and the second insulating layer **15**.
(17) As shown in FIG. **2** and FIG. **3**, the dust collector **1** according to the embodiment can be formed to have, for example, a square shape in planar view. An example in which the dust collector **1** is formed to have the square shape will be explained below; however, the dust collector **1** may have other shapes.
(18) The support base layer **11**, the first insulating layer **13**, the second insulating layer **15**, and the dust attraction layer **16** are formed to have the approximately same square shape and size in planar view in which each side has a length L**1**.
(19) The second electrode layer **14** is formed to have a square shape in which each side has a length L**2** shorter than the length L**1**.
(20) The first electrode layer **12** is formed to have a square shape in which each side has a length L**3** shorter than the length L**2**.
(21) Accordingly, the first electrode layer **12** fits within the second electrode layer **14**, and the second electrode layer **14** fits within the support base layer **11**, the first insulating layer **13**, the second insulating layer **15**, and the dust attraction layer **16** in planar view seen from the dust attraction layer **16** side.
(22) As shown in FIG. **2**, the first electrode layer **12** is formed so that edges of the first electrode layer **12** are located on an inner side by a width W or more from edges of the second electrode layer **14**. The width W is desirably 5 mm or more.
(23) A thickness T**1** of the first electrode layer **12** is thinner than a thickness T**2** of the first insulating layer **13**. A thickness T**3** of the second electrode layer **14** is thinner than a thickness T**4** of the second insulating layer **15**.
(24) As shown in FIG. **3**, each of the support base layer **11**, the first electrode layer **12**, the first insulating layer **13**, the second insulating layer **15**, and the dust attraction layer **16** is formed without having a lacking part in planar view, each with their own unique layer.
(25) On the other hand, the second electrode layer **14** has circular through holes **14**H at equal intervals and piercing therethrough in the thickness direction as lacking parts. An inner diameter of the through holes **14**H is, for example, 5 mm to 15 mm, and an interval between adjacent through holes **14**H is 10 mm to 30 mm.
(26) A second electrode layer **14**A according to a modification example of the second electrode layer **14** has slits **14**S as lacking parts. The slits **14**S are provided alternately from one side of the second electrode layer **14** toward a side opposed to the one side. Therefore, unevenness in dust collection performance can be prevented.
(27) The lacking parts are provided so as not to interfere with energizing the second electrode layer **14**.
(28) FIG. **4** is a cross-sectional view showing a state of electric fields generated in the dust collector **1**. In FIG. **4**, outlined arrows X and outlined arrows Y indicate electric fields.
(29) A positive terminal of the power supply device **2** is connected to one of the first electrode layer **12** and the second electrode layer **14**, and a negative terminal thereof is connected to the other of them to energize.
(30) In a case where particles consist of an insulating material, it is desirable that the second electrode layer **14** is connected to the opposite polarity of which the insulating material tends to be charged (triboelectric effect) from a dust collection efficiency perspective.
(31) The electric field from the second electrode layer **14** passes the second insulating layer **15** and the dust attraction layer **16** from the second electrode layer **14** as shown by the arrows X, reaches a space where particles exist, passes the dust attraction layer **16** and the second insulating layer **15** again, and passes the first insulating layer **13** through the lacking parts of the second electrode layer **14** to reach a balanced part with respect to the electric field from the first electrode layer **12** shown by the arrows Y.
(32) According to the above, an unbalanced electric field is generated in an outer side direction of the dust attraction layer **16** of the dust collector **1**, and then, particles are repelled from a workpiece or the like by the electric field directed to an outer side and particles move along the electric field by the electric field directed to an inner side to be collected by the dust attraction layer **16**. Here, the unbalanced electric field indicates a range excluding an offset point that serves as the balanced part due to cancellation of the electric fields and that has no directionality and refers to an electric field where Coulomb force having directionality is generated. In the embodiment, the first electrode layer **12** is arranged not at the same height position as the second electrode layer **14** in front view of FIG. **4** but lower than the first insulating layer **13** with the first insulating layer **13** interposed therebetween. Accordingly, the balanced part of the electric fields exists not at an upper end of the arrow X of FIG. **4** but at a position where the arrows X face the arrows Y, namely, at the inside of the dust collector **1**; therefore, dust entering the electric field can be moved constantly in a fixed direction shown by the arrows X of FIG. **4** and collected to the dust attraction layer **16** provided on an upper end surface of the dust collector **1**.
(33) Here, the second electrode layer **14** covers the entire surface of the first electrode layer **12** in planar view. Accordingly, all orientations of the electric field from the second electrode layer **14** are aligned in the direction of the lacking parts, namely, the inner direction of the dust collector **1** in planar view; therefore, it is possible to prevent particles from scattering to the outer side of the dust collector **1**.
(34) Furthermore, in a case where particles are made of a conductor or some insulating material, the particles are charged inside the electric field due to electrostatic induction, and in a case where particles are made of an insulating material or a semiconductor, the particles are charged inside the electric field due to dielectric polarization. Particles are charged in both cases.
(35) The charged particles are attracted by the electric field and collected by the dust attraction layer **16**.
(36) Here, experimental results will be explained. Experiments were made using the dust collector **1** according to the embodiment having 60 mm×60 mm.
(37) A voltage of 2 kV was applied to the first electrode layer **12** of the dust collector **1** set up vertically to a floor surface, and a voltage of −4 kV which was a reverse-polarity voltage higher than the voltage applied to the first electrode layer **12** was applied to the second electrode layer **14**.
(38) Samples were hung by strings at intervals on the dust attraction layer **16** side of the dust collector **1**.
(39) In a sample of aluminum foil with 10 mm×0.5 mm, the attracting effect was confirmed within a distance of approximately 100 mm.
(40) In a sample of a silicon wafer with 10 mm×0.5 mm, the attracting effect was confirmed within a distance of approximately 50

mm.
(41) In a sample of resin (acrylic resin) with 10×0.5 mm, the attracting effect was confirmed within a distance of approximately 60 mm.
(42) As described above, the dust collector **1** according to the embodiment includes the first electrode layer **12** formed of the conductor, the second electrode layer **14** formed of the conductor, having sufficient size to cover the first electrode layer **12** in planar view, having the lacking parts piercing therethrough in the thickness direction (the through holes **14**H, the slits **14**S) and arranged to be opposed to the first electrode layer **12**, and the insulating layer (the first insulating layer **13**, the second insulating layer **15**) formed of the insulative material, insulating the first electrode layer **12** and the second electrode layer **14**, and forming a single merged layer structure with the first electrode layer **12** and the second electrode layer **14**.
(43) A dust collection method according to the embodiment includes applying a voltage to the second electrode layer **14** that is arranged in layer so as to be opposed to the first electrode layer **12** with the insulating layer (the first insulating layer **13**) interposed therebetween and that has sufficient size to cover the first electrode layer **12** in planar view, and applying a voltage to the first electrode layer **12**, to thereby cause the electric field generated from the second electrode layer **14** toward the outer side direction to pass through the lacking parts (the through holes **14**H, the slits **14**S) provided in the second electrode layer **14**, generating the unbalanced electric field reaching the first electrode layer **12**, and conveying particles along the unbalanced electric field.
(44) Accordingly, there is an advantage that a dust collector that can be installed even at a small space and that can collect dust can be provided according to the embodiment.
REFERENCE SIGNS LIST
(45) **1**: Dust collector **2**: Power supply device **11**: Support base layer **12**: First electrode layer **13**: First insulating layer **14**: Second electrode layer **14**A: Second electrode layer **14**H: Through hole **14**S: Slit **15**: Second insulating layer **16**: Dust attraction layer

Claims

1. A dust collector comprising: a first electrode layer formed of a conductor; a second electrode layer formed of a conductor, the second electrode layer having through holes penetrating the second electrode layer in a thickness direction, a size of the second electrode layer being larger than a size of the first electrode layer in planar view, and the second electrode layer being arranged to be opposed to the first electrode layer; and an insulating layer formed of an insulative material, insulating the first electrode layer and the second electrode layer, and forming one layer structure with the first electrode layer and the second electrode layer.
2. The dust collector according to claim 1, wherein the first electrode layer is formed so that edges of the first electrode layer are located on an inner side by 5 mm or more from edges of the second electrode layer.
3. The dust collector according to claim 1, further comprising: a dust attraction layer arranged to be opposed to the second electrode layer with the insulating layer interposed therebetween.
4. The dust collector according to claim 1, further comprising: a support base layer arranged to be opposed to the first electrode layer with the insulating layer interposed therebetween.
5. The dust collector according to claim 1, wherein the through holes have a diameter of 5 mm to 15 mm, and an interval between adjacent through holes is 10 mm to 30 mm.
6. The dust collector according to claim 1, wherein the insulating layer is formed of polyimide.
7. A dust collection method comprising: applying a voltage to a second electrode layer that is arranged in layers opposed to a first electrode layer with an insulating layer interposed between the first and second electrode layers, a size of the second electrode layer being larger than a size of the first electrode layer in planar view; and applying a voltage to the first electrode layer, to thereby cause an electric field generated from the second electrode layer toward an outer side direction to pass through holes that penetrate the second electrode layer in a thickness direction, and reach the first electrode layer.
8. The dust collection method according to claim 7, wherein particles are conveyed along the generated electric field.
9. The dust collector according to claim 1, wherein each of the first and second electrode layers has a square shape.